

AFRERA Enterprise Verification & Gap Analysis (EVGA)

Phase 13: Enhanced Evidence-Based Verification

Date: 2026-07-27

Objective: Verify the 205 new enterprise platform capabilities (CAP-076 to CAP-288) identified in Phase 11 against repository evidence to determine actual implementation status.

Executive Summary

All 205 new enterprise platform capabilities identified in Phase 11 have been verified against the repository. The verification confirms that **none of these capabilities are implemented** in the current codebase.

Verification Results:

- Total New Capabilities Verified: 205
 - Capabilities with Backend Evidence: 0 (0%)
 - Capabilities with Database Evidence: 0 (0%)
 - Capabilities with API Evidence: 0 (0%)
 - Capabilities with UI Evidence: 0 (0%)
 - Capabilities with Test Evidence: 0 (0%)
 - **Overall Implementation Status:** 0%
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Verification Methodology

For each of the 205 new capabilities (CAP-076 to CAP-288), the following evidence types were searched:

1. **Backend Code:** JavaScript/Node.js files in `backend/src/`
2. **Database Schema:** SQL files, migration files, schema definitions
3. **API Endpoints:** Route definitions, controller files
4. **UI Components:** React/JSX files in `frontend/src/`

Verification Results by Platform

Platform 1: Multilingual Intelligence Platform (CAP-076 to CAP-085)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-076	Automatic Language Detection	NO	NO	NO	NO	NO	MISSING
CAP-077	Indian Language Support	NO	NO	NO	NO	NO	MISSING
CAP-078	Northeast Language Support	NO	NO	NO	NO	NO	MISSING
CAP-079	Multilingual UI Components	NO	NO	NO	NO	NO	MISSING
CAP-080	Multilingual Content Management	NO	NO	NO	NO	NO	MISSING
CAP-081	Translation Memory	NO	NO	NO	NO	NO	MISSING
CAP-082	Regional Terminology Support	NO	NO	NO	NO	NO	MISSING
CAP-083	Agriculture Glossary	NO	NO	NO	NO	NO	MISSING
CAP-084	Voice Pronunciation	NO	NO	NO	NO	NO	MISSING
CAP-085	Cultural Localization	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/10 capabilities)

Platform 2: Enterprise Conversational AI Platform (CAP-086 to CAP-107)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-086	AI Assistant Framework	NO	NO	NO	NO	NO	MISSING

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-087	Farmer AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-088	Buyer AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-089	Seller AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-090	FPO AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-091	Logistics AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-092	Warehouse AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-093	Finance AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-094	GST AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-095	Insurance AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-096	Dietitian AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-097	Agronomist AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-098	Vet AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-099	Fish Farming AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-100	Hydroponics AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-101	Export AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-102	Government Scheme AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-103	Compliance AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-104	Laboratory AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-105	Nutrition AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-106	Sustainability AI Assistant	NO	NO	NO	NO	NO	MISSING
CAP-107	Customer Support AI Assistant	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/22 capabilities)

Platform 3: Voice AI Platform (CAP-108 to CAP-115)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-108	Speech-to-Text	NO	NO	NO	NO	NO	MISSING
CAP-109	Text-to-Speech	NO	NO	NO	NO	NO	MISSING
CAP-110	Voice Search	NO	NO	NO	NO	NO	MISSING
CAP-111	Voice Ordering	NO	NO	NO	NO	NO	MISSING
CAP-112	Voice Navigation	NO	NO	NO	NO	NO	MISSING
CAP-113	Voice Training	NO	NO	NO	NO	NO	MISSING
CAP-114	Voice Forms	NO	NO	NO	NO	NO	MISSING
CAP-115	Voice Complaint Registration	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/8 capabilities)

Platform 4: Nutrition Intelligence OS (CAP-116 to CAP-131)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-116	Food Composition Database	NO	NO	NO	NO	NO	MISSING
CAP-117	AI Nutrient Calculator	NO	NO	NO	NO	NO	MISSING
CAP-118	Cost-per-Nutrient Engine	NO	NO	NO	NO	NO	MISSING
CAP-119	AI Dietitian Platform	NO	NO	NO	NO	NO	MISSING
CAP-120	AI Nutrition Coach	NO	NO	NO	NO	NO	MISSING
CAP-121	Nutrition Label Generator	NO	NO	NO	NO	NO	MISSING

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-122	Nutrient Traceability	NO	NO	NO	NO	NO	MISSING
CAP-123	Nutrient Verification	NO	NO	NO	NO	NO	MISSING
CAP-124	Nutrient Marketplace	NO	NO	NO	NO	NO	MISSING
CAP-125	Personalized Nutrition	NO	NO	NO	NO	NO	MISSING
CAP-126	Clinical Nutrition	NO	NO	NO	NO	NO	MISSING
CAP-127	Food Safety	NO	NO	NO	NO	NO	MISSING
CAP-128	Allergen Management	NO	NO	NO	NO	NO	MISSING
CAP-129	Health Claim Validation	NO	NO	NO	NO	NO	MISSING
CAP-130	Nutrient Analytics	NO	NO	NO	NO	NO	MISSING
CAP-131	Research & Development	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/16 capabilities)

Platform 5: Laboratory ERP (LIMS) (CAP-132 to CAP-143)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-132	Laboratory Registration	NO	NO	NO	NO	NO	MISSING
CAP-133	NABL/Accredited Lab Management	NO	NO	NO	NO	NO	MISSING
CAP-134	Sample Collection	NO	NO	NO	NO	NO	MISSING
CAP-135	Chain of Custody	NO	NO	NO	NO	NO	MISSING
CAP-136	Sample Tracking	NO	NO	NO	NO	NO	MISSING
CAP-137	Test Scheduling	NO	NO	NO	NO	NO	MISSING
CAP-138	Instrument Integration	NO	NO	NO	NO	NO	MISSING

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-139	Test Reports	NO	NO	NO	NO	NO	MISSING
CAP-140	Digital Signatures	NO	NO	NO	NO	NO	MISSING
CAP-141	Certificate Generation	NO	NO	NO	NO	NO	MISSING
CAP-142	Historical Trends	NO	NO	NO	NO	NO	MISSING
CAP-143	Recall Support	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/12 capabilities)

Platform 6: Northeast Organic Traceability OS (CAP-144 to CAP-171)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-144	Organic Farm Registry	NO	NO	NO	NO	NO	MISSING
CAP-145	Organic Farmer Registry	NO	NO	NO	NO	NO	MISSING
CAP-146	Organic Land Registry	NO	NO	NO	NO	NO	MISSING
CAP-147	Organic Crop Registry	NO	NO	NO	NO	NO	MISSING
CAP-148	Organic Livestock Registry	NO	NO	NO	NO	NO	MISSING
CAP-149	Organic Fisheries Registry	NO	NO	NO	NO	NO	MISSING
CAP-150	Organic Forest Produce Registry	NO	NO	NO	NO	NO	MISSING
CAP-151	Organic Certification Registry	NO	NO	NO	NO	NO	MISSING
CAP-152	Organic Laboratory Registry	NO	NO	NO	NO	NO	MISSING
CAP-153	Organic Collection Registry	NO	NO	NO	NO	NO	MISSING

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-154	Organic Warehouse Registry	NO	NO	NO	NO	NO	MISSING
CAP-155	Organic Processing Registry	NO	NO	NO	NO	NO	MISSING
CAP-156	Organic Packaging Registry	NO	NO	NO	NO	NO	MISSING
CAP-157	Organic Logistics Registry	NO	NO	NO	NO	NO	MISSING
CAP-158	Organic Marketplace Registry	NO	NO	NO	NO	NO	MISSING
CAP-159	Organic Export Registry	NO	NO	NO	NO	NO	MISSING
CAP-160	Organic Consumer Registry	NO	NO	NO	NO	NO	MISSING
CAP-161	Chain of Custody Tracking	NO	NO	NO	NO	NO	MISSING
CAP-162	Input Traceability	NO	NO	NO	NO	NO	MISSING
CAP-163	Harvest Traceability	NO	NO	NO	NO	NO	MISSING
CAP-164	Logistics Traceability	NO	NO	NO	NO	NO	MISSING
CAP-165	Cold Chain Traceability	NO	NO	NO	NO	NO	MISSING
CAP-166	Processing Traceability	NO	NO	NO	NO	NO	MISSING
CAP-167	Consumer Transparency (QR)	NO	NO	NO	NO	NO	MISSING
CAP-168	Organic Fraud Detection	NO	NO	NO	NO	NO	MISSING
CAP-169	Organic Compliance Prediction	NO	NO	NO	NO	NO	MISSING
CAP-170	Certification Risk Scoring	NO	NO	NO	NO	NO	MISSING

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-171	Counterfeit Detection	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/28 capabilities)

Platform 7: GI Intelligence Platform (CAP-172 to CAP-180)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-172	GI Registry	NO	NO	NO	NO	NO	MISSING
CAP-173	GI Verification	NO	NO	NO	NO	NO	MISSING
CAP-174	GI Certification	NO	NO	NO	NO	NO	MISSING
CAP-175	GI Mapping	NO	NO	NO	NO	NO	MISSING
CAP-176	GI Marketplace	NO	NO	NO	NO	NO	MISSING
CAP-177	GI Analytics	NO	NO	NO	NO	NO	MISSING
CAP-178	GI Export	NO	NO	NO	NO	NO	MISSING
CAP-179	GI Premium Engine	NO	NO	NO	NO	NO	MISSING
CAP-180	GI Authenticity	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/9 capabilities)

Platform 8: Food Intelligence OS (CAP-181 to CAP-192)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-181	Food Knowledge Base	NO	NO	NO	NO	NO	MISSING
CAP-182	Taste Profile	NO	NO	NO	NO	NO	MISSING
CAP-183	Texture Profile	NO	NO	NO	NO	NO	MISSING
CAP-184	Flavor Profile	NO	NO	NO	NO	NO	MISSING

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-185	Cooking Methods Database	NO	NO	NO	NO	NO	MISSING
CAP-186	Traditional Uses Database	NO	NO	NO	NO	NO	MISSING
CAP-187	Medicinal Uses Database	NO	NO	NO	NO	NO	MISSING
CAP-188	Recipes Database	NO	NO	NO	NO	NO	MISSING
CAP-189	Seasonality Database	NO	NO	NO	NO	NO	MISSING
CAP-190	Compatibility Database	NO	NO	NO	NO	NO	MISSING
CAP-191	Storage Guidelines	NO	NO	NO	NO	NO	MISSING
CAP-192	Shelf-Life Database	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/12 capabilities)

Platform 9: Value-Based Commerce OS (CAP-193 to CAP-200)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-193	Quality-Based Pricing	NO	NO	NO	NO	NO	MISSING
CAP-194	Nutrition-Based Pricing	NO	NO	NO	NO	NO	MISSING
CAP-195	Scarcity-Based Pricing	NO	NO	NO	NO	NO	MISSING
CAP-196	GI Premium Pricing	NO	NO	NO	NO	NO	MISSING
CAP-197	Organic Premium Pricing	NO	NO	NO	NO	NO	MISSING
CAP-198	Seasonality-Based Pricing	NO	NO	NO	NO	NO	MISSING
CAP-199	Demand-Based Pricing	NO	NO	NO	NO	NO	MISSING

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-200	AI Value-Based Pricing Engine	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/8 capabilities)

Platform 10: Consumer Health Platform (CAP-201 to CAP-208)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-201	Family Nutrition	NO	NO	NO	NO	NO	MISSING
CAP-202	Disease Management	NO	NO	NO	NO	NO	MISSING
CAP-203	AI Dietician	NO	NO	NO	NO	NO	MISSING
CAP-204	Health Dashboard	NO	NO	NO	NO	NO	MISSING
CAP-205	Allergy Management	NO	NO	NO	NO	NO	MISSING
CAP-206	Nutrient Tracking	NO	NO	NO	NO	NO	MISSING
CAP-207	Meal Planning	NO	NO	NO	NO	NO	MISSING
CAP-208	Shopping Recommendations	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/8 capabilities)

Platform 11: Indigenous Knowledge Platform (CAP-209 to CAP-216)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-209	Traditional Recipes Database	NO	NO	NO	NO	NO	MISSING
CAP-210	Traditional Medicine Database	NO	NO	NO	NO	NO	MISSING

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-211	Indigenous Farming Database	NO	NO	NO	NO	NO	MISSING
CAP-212	Oral History Database	NO	NO	NO	NO	NO	MISSING
CAP-213	Tribal Knowledge Database	NO	NO	NO	NO	NO	MISSING
CAP-214	Documentation System	NO	NO	NO	NO	NO	MISSING
CAP-215	Protection System	NO	NO	NO	NO	NO	MISSING
CAP-216	IP Management	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/8 capabilities)

Platform 12: Biodiversity Intelligence (CAP-217 to CAP-223)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-217	Species Database	NO	NO	NO	NO	NO	MISSING
CAP-218	Native Crops Database	NO	NO	NO	NO	NO	MISSING
CAP-219	Traditional Varieties Database	NO	NO	NO	NO	NO	MISSING
CAP-220	Medicinal Plants Database	NO	NO	NO	NO	NO	MISSING
CAP-221	Wild Foods Database	NO	NO	NO	NO	NO	MISSING
CAP-222	Conservation Tracking	NO	NO	NO	NO	NO	MISSING
CAP-223	AI Risk Prediction	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/7 capabilities)

Platform 13: AI Copilot Framework (CAP-224 to CAP-230)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-224	Finance Copilot	NO	NO	NO	NO	NO	MISSING
CAP-225	Logistics Copilot	NO	NO	NO	NO	NO	MISSING
CAP-226	Warehouse Copilot	NO	NO	NO	NO	NO	MISSING
CAP-227	Insurance Copilot	NO	NO	NO	NO	NO	MISSING
CAP-228	Nutrition Copilot	NO	NO	NO	NO	NO	MISSING
CAP-229	Marketplace Copilot	NO	NO	NO	NO	NO	MISSING
CAP-230	Copilot Framework	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/7 capabilities)

Platform 14: Knowledge Graph Platform (CAP-231 to CAP-235)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-231	Entity Relationship Graph	NO	NO	NO	NO	NO	MISSING
CAP-232	Graph Database	NO	NO	NO	NO	NO	MISSING
CAP-233	Graph Query Engine	NO	NO	NO	NO	NO	MISSING
CAP-234	Graph Visualization	NO	NO	NO	NO	NO	MISSING
CAP-235	Graph Analytics	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/5 capabilities)

Platform 15: Omnichannel AI Platform (CAP-236 to CAP-246)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-236	Web AI Integration	NO	NO	NO	NO	NO	MISSING
CAP-237	Android AI Integration	NO	NO	NO	NO	NO	MISSING

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-238	iOS AI Integration	NO	NO	NO	NO	NO	MISSING
CAP-239	WhatsApp AI Integration	NO	NO	NO	NO	NO	MISSING
CAP-240	SMS AI Integration	NO	NO	NO	NO	NO	MISSING
CAP-241	Telegram AI Integration	NO	NO	NO	NO	NO	MISSING
CAP-242	Email AI Integration	NO	NO	NO	NO	NO	MISSING
CAP-243	Voice AI Integration	NO	NO	NO	NO	NO	MISSING
CAP-244	IVR AI Integration	NO	NO	NO	NO	NO	MISSING
CAP-245	Kiosk AI Integration	NO	NO	NO	NO	NO	MISSING
CAP-246	Omnichannel Orchestration	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/11 capabilities)

Platform 16: Food Safety ERP (CAP-247 to CAP-254)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-247	HACCP Management	NO	NO	NO	NO	NO	MISSING
CAP-248	FSSAI Compliance	NO	NO	NO	NO	NO	MISSING
CAP-249	ISO 22000 Compliance	NO	NO	NO	NO	NO	MISSING
CAP-250	Recall Management	NO	NO	NO	NO	NO	MISSING
CAP-251	CAPA Management	NO	NO	NO	NO	NO	MISSING
CAP-252	Food Safety Audit	NO	NO	NO	NO	NO	MISSING
CAP-253	Risk Assessment	NO	NO	NO	NO	NO	MISSING
CAP-254	Corrective Actions	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/8 capabilities)

Platform 17: Shelf-Life Intelligence (CAP-255 to CAP-261)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-255	Temperature Monitoring	NO	NO	NO	NO	NO	MISSING
CAP-256	Humidity Monitoring	NO	NO	NO	NO	NO	MISSING
CAP-257	Packaging Analysis	NO	NO	NO	NO	NO	MISSING
CAP-258	Transport Analysis	NO	NO	NO	NO	NO	MISSING
CAP-259	Storage Analysis	NO	NO	NO	NO	NO	MISSING
CAP-260	Remaining Shelf Life Prediction	NO	NO	NO	NO	NO	MISSING
CAP-261	Spoilage Risk Prediction	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/7 capabilities)

Platform 18: Institutional Procurement ERP (CAP-262 to CAP-268)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-262	Tender Management	NO	NO	NO	NO	NO	MISSING
CAP-263	Demand Forecasting	NO	NO	NO	NO	NO	MISSING
CAP-264	Institution Menu Planning	NO	NO	NO	NO	NO	MISSING
CAP-265	Nutrition Compliance	NO	NO	NO	NO	NO	MISSING
CAP-266	Supply Contracts	NO	NO	NO	NO	NO	MISSING
CAP-267	Quality Inspection	NO	NO	NO	NO	NO	MISSING
CAP-268	Settlement Management	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/7 capabilities)

Platform 19: Digital Product Passport (CAP-269 to CAP-280)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-269	Unique Product ID	NO	NO	NO	NO	NO	MISSING
CAP-270	Lot/Batch Tracking	NO	NO	NO	NO	NO	MISSING
CAP-271	Farm Information	NO	NO	NO	NO	NO	MISSING
CAP-272	Farmer Information	NO	NO	NO	NO	NO	MISSING
CAP-273	Certification Information	NO	NO	NO	NO	NO	MISSING
CAP-274	Processing History	NO	NO	NO	NO	NO	MISSING
CAP-275	Logistics History	NO	NO	NO	NO	NO	MISSING
CAP-276	Sustainability Data	NO	NO	NO	NO	NO	MISSING
CAP-277	Carbon Data	NO	NO	NO	NO	NO	MISSING
CAP-278	Quality Reports	NO	NO	NO	NO	NO	MISSING
CAP-279	Recall Status	NO	NO	NO	NO	NO	MISSING
CAP-280	QR Code Generation	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/12 capabilities)

Platform 20: Recipe Intelligence (CAP-281 to CAP-288)

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-281	Recipe Database	NO	NO	NO	NO	NO	MISSING
CAP-282	AI Recipe Generator	NO	NO	NO	NO	NO	MISSING
CAP-283	Nutrition Calculation	NO	NO	NO	NO	NO	MISSING

Capability ID	Capability Name	Backend	Database	API	UI	Test	Overall Status
CAP-284	Ingredient Substitution	NO	NO	NO	NO	NO	MISSING
CAP-285	Cost Calculator	NO	NO	NO	NO	NO	MISSING
CAP-286	Seasonal Recipes	NO	NO	NO	NO	NO	MISSING
CAP-287	Regional Recipes	NO	NO	NO	NO	NO	MISSING
CAP-288	Institutional Recipes	NO	NO	NO	NO	NO	MISSING

Platform Status: 0% implemented (0/8 capabilities)

Summary of Verification Results

Overall Statistics

Metric	Count	Percentage
Total Capabilities Verified	205	100%
Capabilities with Any Evidence	0	0%
Capabilities with Backend Evidence	0	0%
Capabilities with Database Evidence	0	0%
Capabilities with API Evidence	0	0%
Capabilities with UI Evidence	0	0%
Capabilities with Test Evidence	0	0%
Capabilities Missing	205	100%

Platform-Level Summary

Platform	Capabilities	Implemented	Percentage
Multilingual Intelligence Platform	10	0	0%

Platform	Capabilities	Implemented	Percentage
Enterprise Conversational AI Platform	22	0	0%
Voice AI Platform	8	0	0%
Nutrition Intelligence OS	16	0	0%
Laboratory ERP (LIMS)	12	0	0%
Northeast Organic Traceability OS	28	0	0%
GI Intelligence Platform	9	0	0%
Food Intelligence OS	12	0	0%
Value-Based Commerce OS	8	0	0%
Consumer Health Platform	8	0	0%
Indigenous Knowledge Platform	8	0	0%
Biodiversity Intelligence	7	0	0%
AI Copilot Framework	7	0	0%
Knowledge Graph Platform	5	0	0%
Omnichannel AI Platform	11	0	0%
Food Safety ERP	8	0	0%
Shelf-Life Intelligence	7	0	0%
Institutional Procurement ERP	7	0	0%
Digital Product Passport	12	0	0%
Recipe Intelligence	8	0	0%
TOTAL	205	0	0%

Key Findings

1. **Complete Absence of New Enterprise Platforms:** None of the 205 new capabilities have any implementation evidence in the repository.

2. **No Backend Infrastructure:** No backend services, controllers, or business logic exist for any of the new platforms.
 3. **No Database Schema:** No database tables, migrations, or schema definitions exist for any of the new platforms.
 4. **No API Endpoints:** No API routes or endpoints exist for any of the new platforms.
 5. **No UI Components:** No frontend components or pages exist for any of the new platforms.
 6. **No Test Coverage:** No test files exist for any of the new platforms.
 7. **Critical Gap:** The absence of these platforms represents a fundamental gap in AFRERA's evolution from an Agriculture ERP to a Food Intelligence + Rural Economy Operating System.
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Recommendations

Immediate Actions

1. **Prioritize Multilingual Intelligence Platform** - Critical for Northeast adoption and national scalability.
2. **Prioritize Northeast Organic Traceability OS** - Critical for organic certification and premium product verification.
3. **Prioritize Nutrition Intelligence OS** - Critical for value-based commerce differentiation.
4. **Prioritize Enterprise Conversational AI Platform** - Critical for user experience and accessibility.
5. **Prioritize Laboratory ERP (LIMS)** - Critical for nutrient verification and quality assurance.

Architecture Recommendations

1. **Create New Enterprise Platform Layer** - These platforms should be first-class capabilities, not submodules.
2. **Implement GS1 Standards** - Use GS1 identifiers, Critical Tracking Events (CTEs), and Key Data Elements (KDEs) for traceability.
3. **Implement Digital Product Passport** - Persistent digital identity for all products.

4. **Implement Knowledge Graph** - Connect entities across the ecosystem.

5. **Implement Omnichannel AI** - Unified AI across all channels.

Technology Recommendations

1. **Multilingual AI** - Use Google Cloud Translation, Microsoft Translator, or similar.

2. **Conversational AI** - Use RAG-based architecture with vector databases.

3. **Voice AI** - Use speech-to-text and text-to-speech APIs.

4. **Graph Database** - Use Neo4j or similar for knowledge graph.

5. **Laboratory Integration** - Use LIMS standards and instrument APIs.

Next Phase

The next phase (Phase 14) will update the heat maps with the new domains and capabilities to provide a visual maturity assessment.

Phase 13 Status: COMPLETED

Total New Capabilities Verified: 205

Capabilities with Evidence: 0 (0%)

Capabilities Missing: 205 (100%)

Overall Implementation Status: 0%